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Python Modeling with GR4J

← </build_future>

Motivation & Objectives

Problem Context

- Many watersheds lack reliable streamflow records, making runoff predictions difficult.
- Poor prediction limits flood-risk assessment, infrastructure design, and water-resources planning.
- The open challenge is parameters estimation when no local streamflow data exist for calibration.
- Conceptual models such as GR4J are widely used because they need limited input data yet perform well.

Research Question

How can GR4J model parameters be reliably estimated for ungauged watersheds using regionalization techniques to improve streamflow prediction in data-scarce regions?

Objectives

The main objectives for modeling GR4J:

- Develop a python code to automate calibration of rainfall-runoff models
- Apply the GR4J model across approximately 667 watersheds located in the U.S. from 1980 to 2014
- Evaluate performance of model using calibration and validation results
- Use U.S. catchments for future application to Great Britain catchments

GR4J Model

What is the GR4J Model?

- GR4J stands for a four-parameter daily rainfall-runoff model
- Inputs are daily precipitation and evapotranspiration ; output is streamflow/discharge
- The model represents watershed behavior through storage exchange routing and timing
- Its simplicity makes it useful for both gauged and ungauged catchments

```python
from pathlib import Path
import pandas as pd
import datetime
from hydrogr import InputDataHandler, ModelGr4j
import plotly.graph_objects as go
```

```python
data_path = Path.cwd().parent / 'data'
df = pd.read_pickle(data_path / 'L0123001.pkl')
df.columns = ['date', 'precipitation', 'temperature', 'evapotranspiration', 'flow', 'flow_mm']
df.index = df['date']

inputs = InputDataHandler(ModelGr4j, df)
start_date = datetime.datetime(1989, 1, 1, 0, 0)
end_date = datetime.datetime(1999, 12, 31, 0, 0)
inputs = inputs.get_sub_period(start_date, end_date)
```

```python
# Set the model :
parameters = {"X1": 257.238, "X2": 1.012, "X3": 88.235, "X4": 2.208}

model = ModelGr4j(parameters)
model.set_parameters(parameters)  # Re-define the parameters for demonstration purpose.

# Initial state :
initial_states = {
    "production_store": 0.5,
    "routing_store": 0.6,
    "uh1": None,
    "uh2": None
}
model.set_states(initial_states)
```

hydrogr/example/gr4j.ipynb at master · SimonDelmas/hydrogr · GitHub

GR4J-model-structure-and-calibrated-parameters-Perrin-et-al-2003-P-is-precipitation.png (823×1155)

X1

Production store capacity

X2

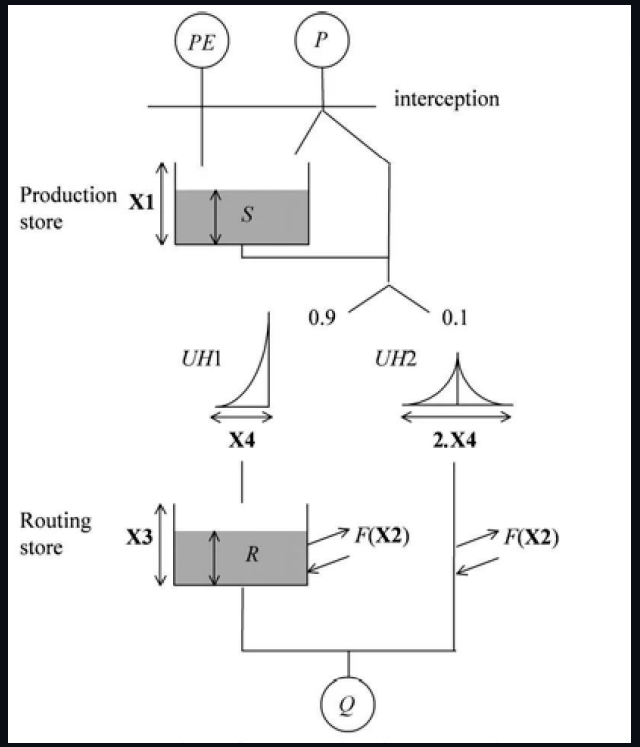
Groundwater exchange

X3

Routing store capacity

X4

Unit hydrograph time base



GR4J Schematic

How Does GR4J Work?

Overall Methodology

Input catchment climate and streamflow data → Calibrate GR4J to obtain best-fit parameters → Validate on separate time period → Run GR4J and evaluate performance using NSE

Calibration & Validation Setup

Calibration & Validation

1980 - 2004

2005 - 2014

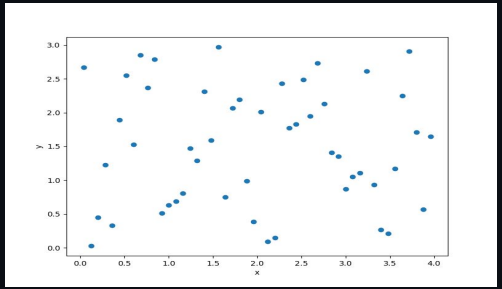
Calibration

Parameter Search for Best Fit

Validation

Test Parameters on a New Time Period

SpotPY + Latin Hypercube Sampling (LHS)



https://smt.readthedocs.io/en/latest/_src_docs/sampling_methods/lhs.html

Number of Repetitions

- More repetitions = more parameter sets explored
- Drawback = Longer Runtime

Nash-Sutcliffe Efficiency (NSE)

Performance Evaluation	NSE
Very good	$0.75 < \text{NSE} \leq 1.00$
Good	$0.65 < \text{NSE} \leq 0.75$
Satisfactory	$0.5 < \text{NSE} \leq 0.65$
Unsatisfactory	$\text{NSE} \leq 0.50$

https://www.researchgate.net/figure/The-Nash-Sutcliffe-Efficiency-Coefficient-values_tbl1_364346865

Linear Regression

- A simple predictive model that estimates an output by fitting a straight-line relationship between input variables and the target value.

Random Forest

- A machine learning method that combines many decision trees to make more accurate and stable predictions than a single tree.

Descriptor Based KNN (K Nearest Neighbors)

- A similarity-based method that uses catchment descriptors to find the **K** most similar catchments and predict values based on their known information.

Support Vector Regression

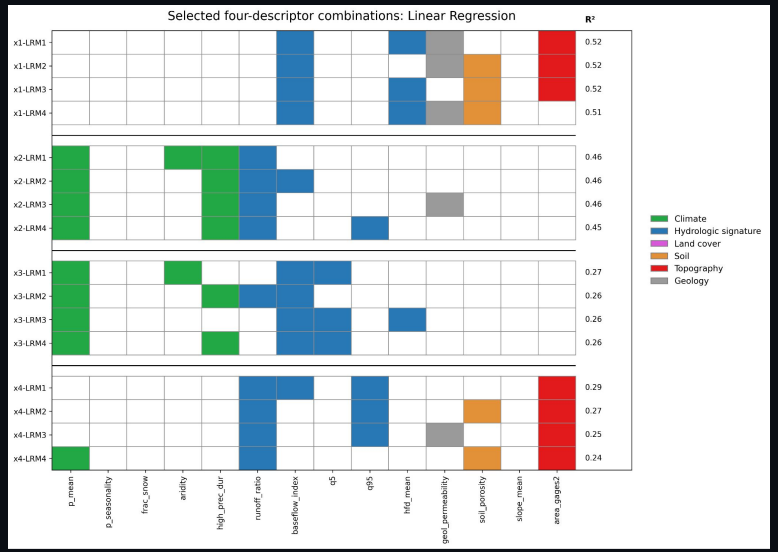
- A machine learning method that predicts GR4J parameters for ungauged catchments by learning relationships between catchment descriptors and calibrated parameters from gauged catchments.

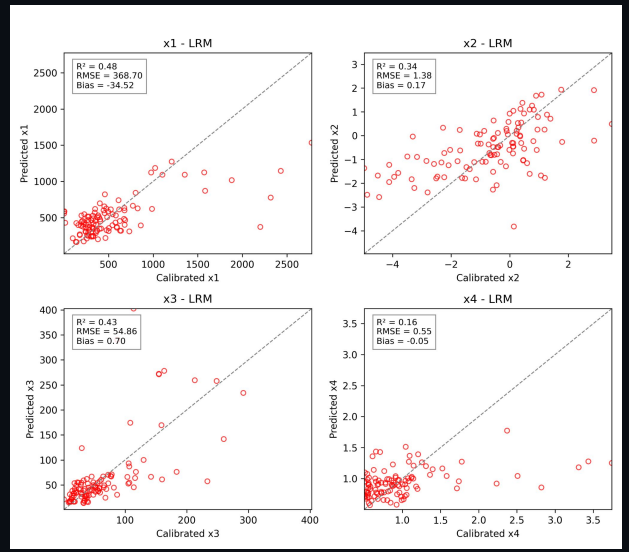
Spatial Proximity

- A regionalization method that estimates GR4J parameters for an ungauged catchment using nearby gauged catchments, based on the assumption that catchments close together often behave similarly.

Machine Learning Methods

Linear Regression Results





R^2 Range Interpretation

$R^2 > 0.85$ very good / excellent

$0.75 < R^2 \leq 0.85$ good

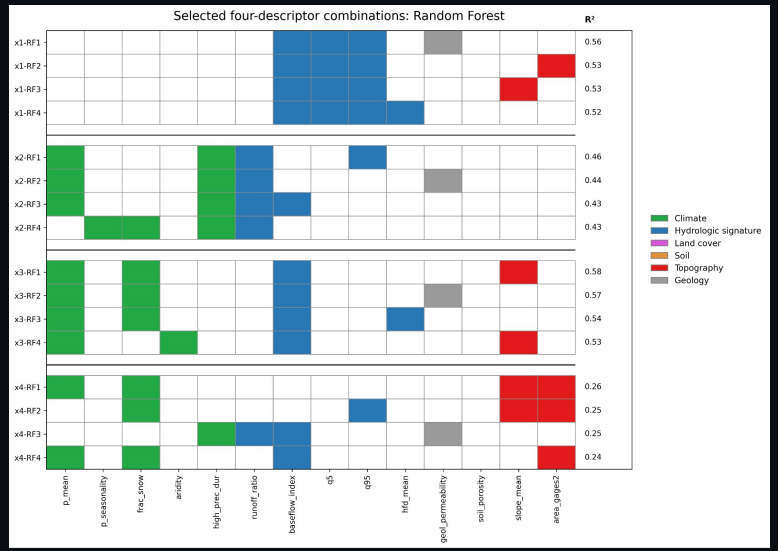
$0.60 < R^2 \leq 0.75$ moderate / satisfactory

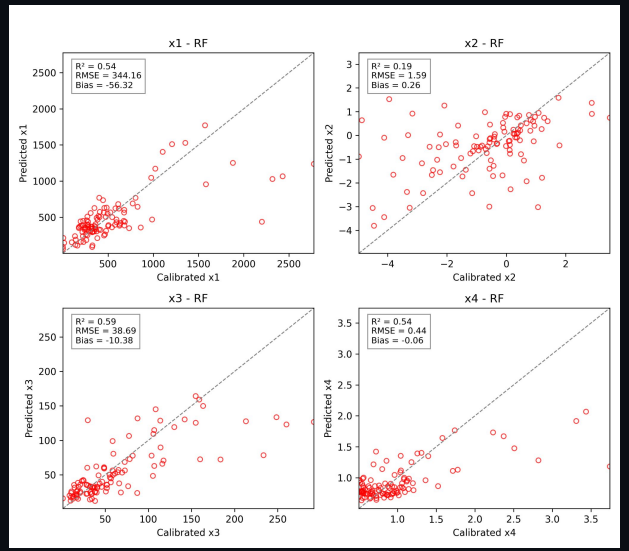
$0.40 < R^2 \leq 0.60$ fair / weak

$R^2 \leq 0.40$ poor

(Moriasi et al., 2015)

Random Forest Results





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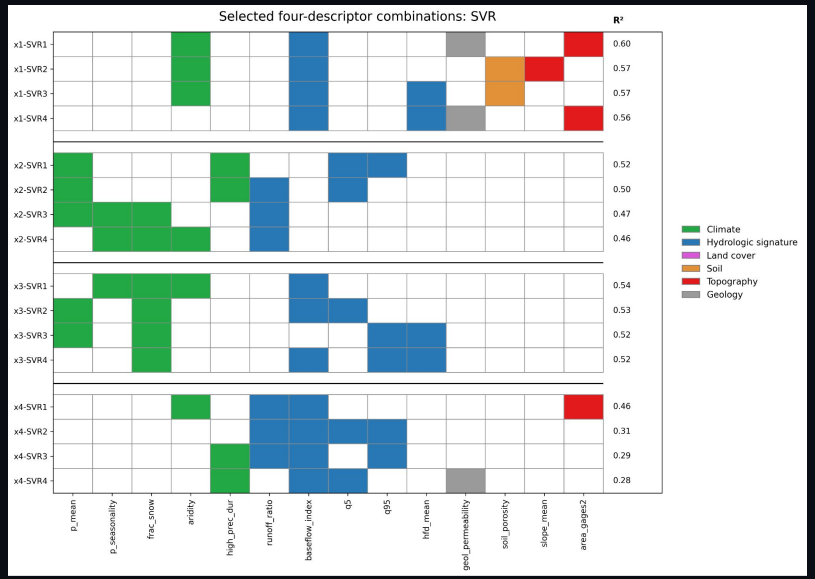
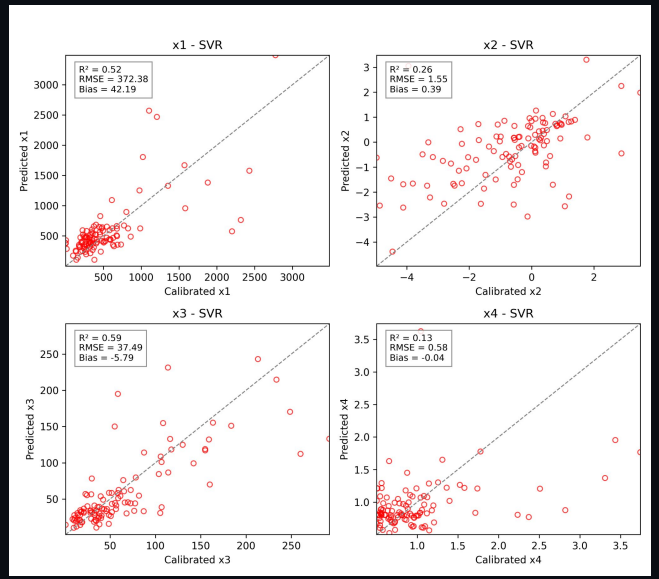
Support Vector Regression Results





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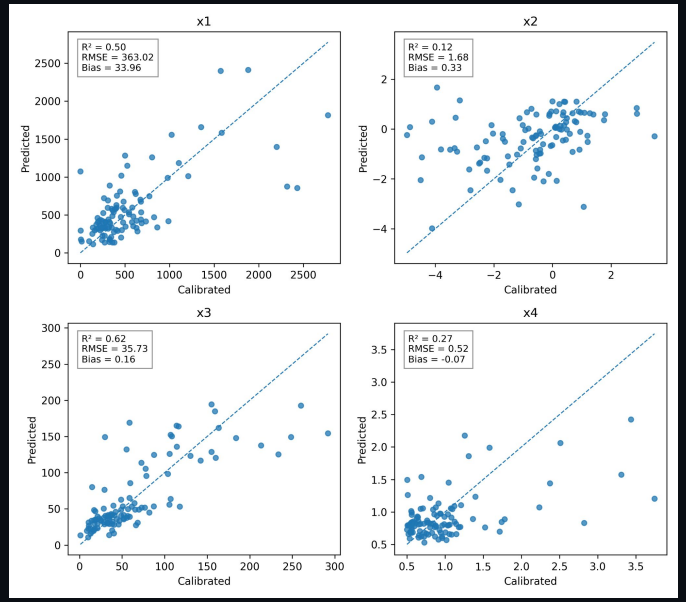
Descriptor Based KNN Results



- Several K values tested which include K = 1, 3, 5, 7, 10, 15
- K = 3 gave the highest average validation performance across X1, X2, X3, X4
- Overall very poor performance for X2 and X4 but weak performance for X1 and moderate for X3

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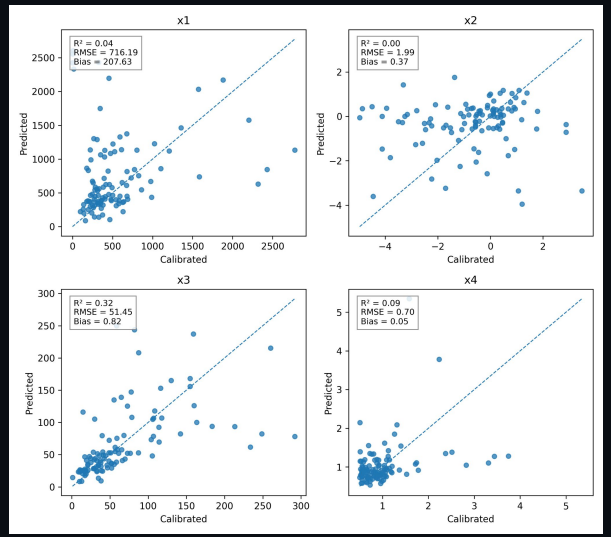
Spatial Proximity Results



- Several K values tested which include K = 1, 3, 5, 7, 10, 15
- K = 5 gave the highest average validation performance across X1, X2, X3, X4
- Overall very poor performance as indicated by the R-Square values for each parameter

R^2 Range Interpretation

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$0.40 < R^2 \leq 0.60$ fair / weak
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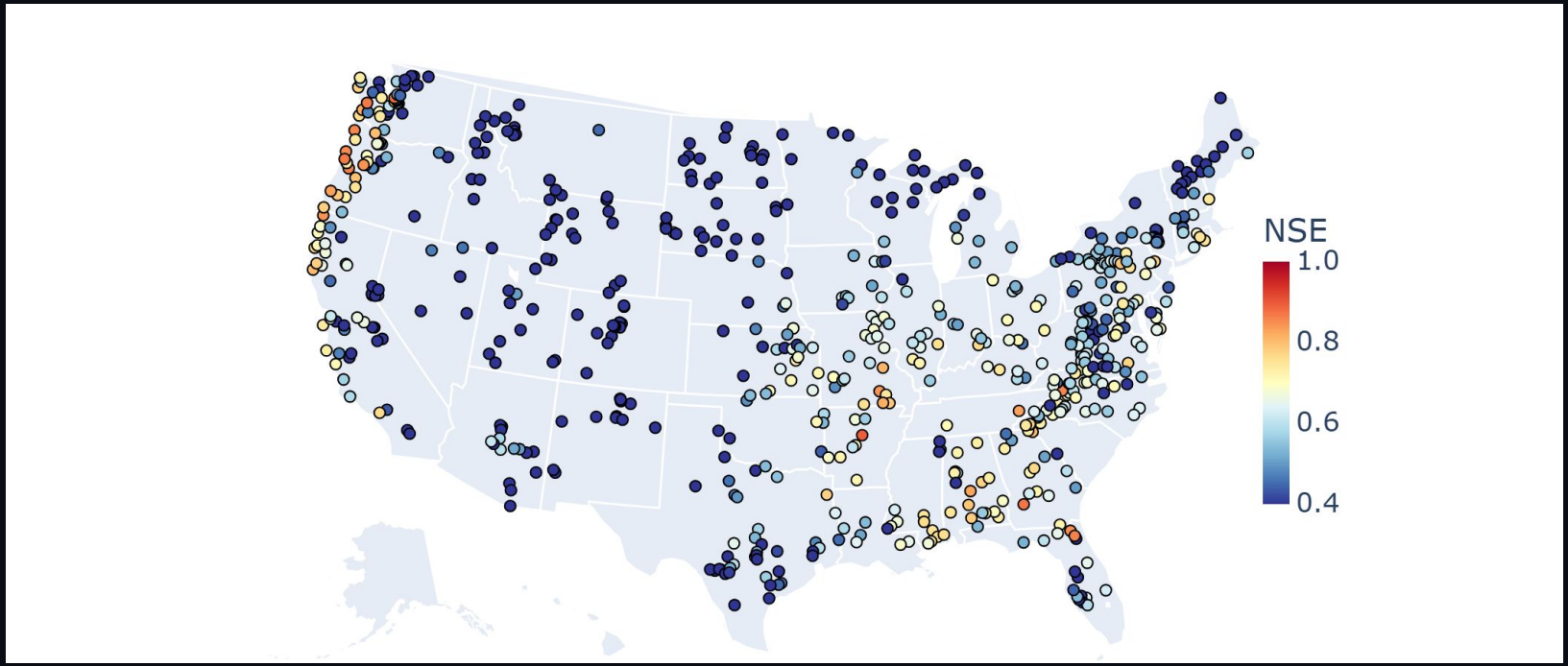
(Moriasi et al., 2015)

Conclusion & NSE Values of 667 U.S. Catchments



GR4J Catchments NSE Values for U.S.

Next Steps

- Find methods to improve initial calibration
- Test and research different algorithms
- Improve code for regionalization methods
- Objective: R-Squared value of 0.80 higher before parameter transfer to another continent or country

	MC	LHS	MLE	MCMC	SCE-UA	SA	DE-MC$_Z$	ROPE
Suited to investigate parameter uncertainty	✓	✓		✓			✓	✓
Allows considering multiple objective functions	✓	✓						
Possible to test prior parameter distributions	✓		(✓)[a]	(✓)[a]	(✓)[a]		(✓)[a]	(✓)[a]
Default algorithm-settings are all-round suited	✓	✓	✓	✓				✓
Suited for parallel computing	✓	✓			(✓)[b]		(✓)[b]	✓
Algorithm learns during sampling			✓	✓	✓	✓	✓	✓

Checked fields indicate positive answers, fields with brackets are partly positive.

[a] Only true during warm-up/burn-in

[b] Only true up to the number of used chains/complexes. They are separated on different CPU cores.

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https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0145180

Thank you

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